AI IN RESEARCH

Refining questions, grounding outlines, and building search strings — with AI.

Political Science Research Workshop • 90 minutes • Hands-on

OPENING

What you'll walk out with

1

A refined research question

Move from a broad interest to a sharp, arguable question with named sub-questions.

2

An outline grounded in real sources

Built in NotebookLM, with citations you can click back to actual passages.

3

A working search string for EDS

Boolean syntax drafted by Gemini, refined live against EBSCO Discovery Service.

FRAMING

AI accelerates research — it doesn't replace it



Where AI helps

- Brainstorming and scoping
- Sub-question generation
- Vocabulary and synonyms
- Draft outlines and search strings

Where AI doesn't replace you

- Picking the angle — you choose, not Gemini
- Verifying every source it suggests
- Evaluating quality of results
- Doing the actual thinking and writing

FRAMING

Still predicting, now working in a loop

Ask three generations of AI to find sources on autonomous weapons systems.

Prediction model

Writes the most likely looking answer in one pass.

Risk. Invents citations that sound real.

Chat assistant

Follows instructions and answers in one try, sometimes after a quick search.

Risk. Does not check its own work.

Reasoning agent

Gemini Deep Research. Plans, searches, reads, hits dead ends, revises its search, and tries again.

Risk. Can still misread or trust a weak source.

Like walking a maze instead of guessing the whole path. **The loop adds persistence, not truth, so the cautions still apply.**

FRAMING

Three cautions to carry into every AI session

1

Unverified citations

General LLMs invent sources that look real. Always click through to confirm the source exists and says what was claimed.

2

Hallucinated facts

Confident-sounding answers can be flat wrong. Cross-check claims against actual sources, especially for cases, statutes, and dates.

3

Privacy and uploads

Don't upload copyrighted PDFs you don't own, unpublished work, personal data, or anything sensitive — even into 'private' notebooks.

PHASE 1 • GEMINI

Refine your question with four prompts

Move from a broad interest to a researchable, arguable question with named sub-questions and key vocabulary.

1 Clarify scope

What are 4–5 different angles a political science paper could take on this topic?

2 Stress-test it

What are the strongest counter-arguments to my thesis?

3 Sub-questions

Give me 5 sub-questions I'd need to answer to argue this well.

4 Vocabulary

List key terms, synonyms, and disciplinary jargon, grouped by concept.

Watch out: if Gemini names a scholar or case, do not trust it yet — that's a Phase 2 job.

PHASE 2 • NOTEBOOKLM

Build your notebook together

We build in parallel. I narrate; you follow on your own topic.

1 Create the notebook

Name it with your refined question.

2 Talk about good sources

Peer-reviewed, policy reports, book chapters. Not blogs.

3 Add 3–5 seed sources

Narrate why each one earns a spot.

4 Generate a Briefing Doc

Synthesis grounded in your sources.

5 Ask an analytical question

Click an inline citation — read the actual passage.

6 Generate an outline

Five sections, each grounded in your sources.

Privacy: only upload sources you have the right to use.

CRITICAL USE

Why grounded beats general for synthesis

NotebookLM

Synthesis tool

- Cites only what's in your notebook
- Inline links to real passages
- Cannot make up sources
- Needs you to bring the sources

Gemini (general LLM)

Ideation tool

- Trained on everything; cites from memory
- May invent plausible-looking citations
- Great for angles, vocabulary, drafts
- Verify every source it names

Use the grounded tool to synthesize. Use the general tool to ideate.

PHASE 3 • GEMINI → EDS

Vocabulary into Boolean search strings

EXAMPLE

```
("autonomous weapon*" OR LAWS
   OR "killer robot*")
AND
("international humanitarian law"
   OR IHL OR "laws of war")
AND
(ethic* OR accountability
   OR "meaningful human control")
```

Iteration moves

Too many results

Add another concept with AND, or narrow with a subject heading.

Too few results

Broaden a synonym group, drop a concept, or add truncation (*).

Use EDS-native tools

Subject term suggestions, field tags (SU, TI, AB).

AI drafts the string. You refine it inside the database.

THE WORKFLOW

Gemini to refine → NotebookLM to ground → EDS to find → loop back.

1

Verify every citation

AI suggests; you confirm.

2

Match tool to job

Grounded for synthesis. General for ideation.

3

Be deliberate uploading

Only what you have a right to use.

Questions? Stop by office hours or email your librarian.